

Private rank 1 came from changing the decision path

Evidence-first selective inference for Olikbochon

0.982

private macro F1

#1

private leaderboard

0.967

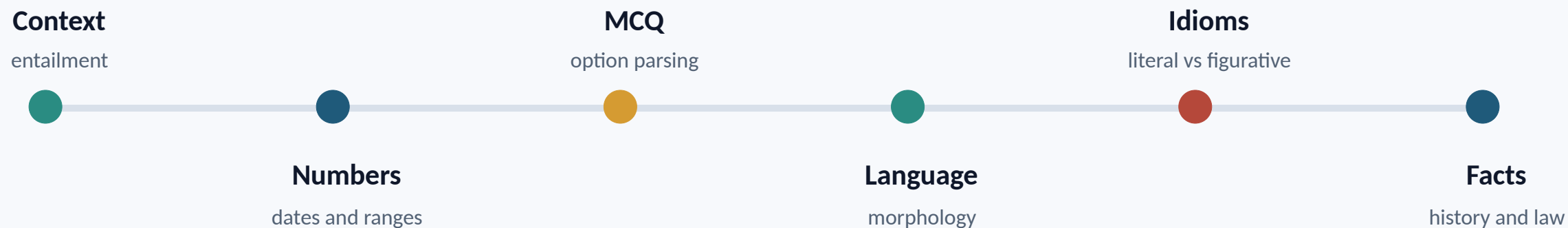
public macro F1

Touhidul Alam Seyam | MD. Abtahee Kabir | Joyeta Barua Moni | Noore Tamanna Orny

BGC Trust University Bangladesh | Chittagong University of Engineering & Technology

One binary label hid many different reasoning tasks

The same label had to cover context faithfulness, language knowledge, and outside facts.

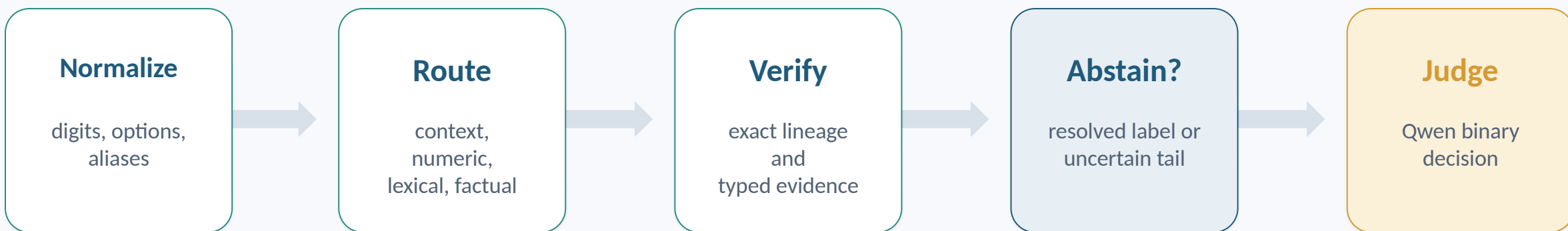


A single judge prompt treated all six as the same problem.

That worked as a baseline, but it was prompt-sensitive and weak on culturally specific relations.

The breakthrough was a selective cascade

Every stage could answer, or abstain. Abstention was a normal outcome, not a failure.

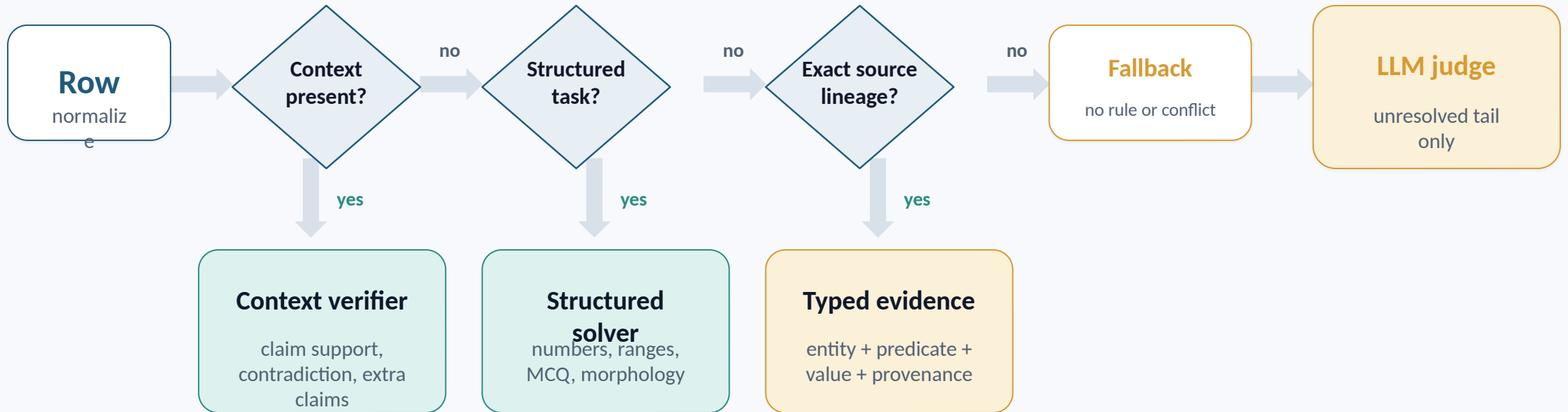


Deterministic stages record the source family and relation behind each decision.

If those fields are not defensible, the row moves forward unchanged.

The router selected an evidence policy before predicting

A branch could return 0, 1, or abstain. Only abstentions moved to the next branch.



Decision contract

branch(row) -> {0, 1, null}; null means abstain, never a guessed label

Deterministic routing handled 93.7% of Phase 1

Only 158 of 2,516 rows reached the Qwen judge.



2,358 deterministic

158

93.72%

resolved before LLM inference

6.28%

unique rows sent to the judge

74.5%

judge-queue reduction from 620

These are unique rows. Some no-context rows used more than one judge view.

Evidence could decide only when the relation matched

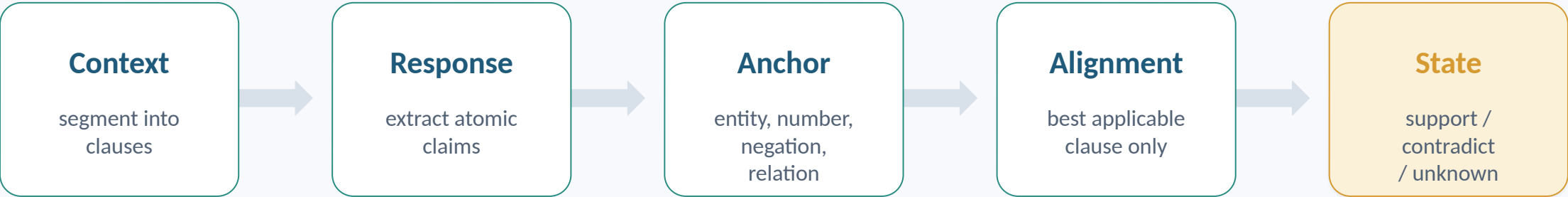
Similarity proposed candidates. It never overruled stronger evidence.

1	Context contradiction	same clause, prompt anchors, no competing claim
2	Exact source lineage	prompt, response, and applicable context agree
3	Structured solver	parsed value, entity, and predicate align
4	Typed evidence	exact idiom sense, law section, or two-source fact
5	Fuzzy retrieval	candidate only; otherwise abstain

Conflict between equal-strength sources also produced abstention.

Context rows were reduced to claim-level entailment tests

The verifier compared atomic response claims against the relevant context clause, not against global word overlap.



Decision rule over response claims $c_1 \dots c_n$

Any contradiction	exists i : $\text{state}(c_i) = \text{contradict}$	label 0
Complete support	for all i : $\text{state}(c_i) = \text{support}$; no extra factual claim	label 1
Incomplete alignment	otherwise	abstain

Qwen judged 158 rows, not the whole test set

The fallback was short, deterministic, and biased toward catching hallucinations.

Context row

1 entailment view

Compare the response against the supplied passage.

Any contradiction returns 0.

No-context row

Up to 3 short views

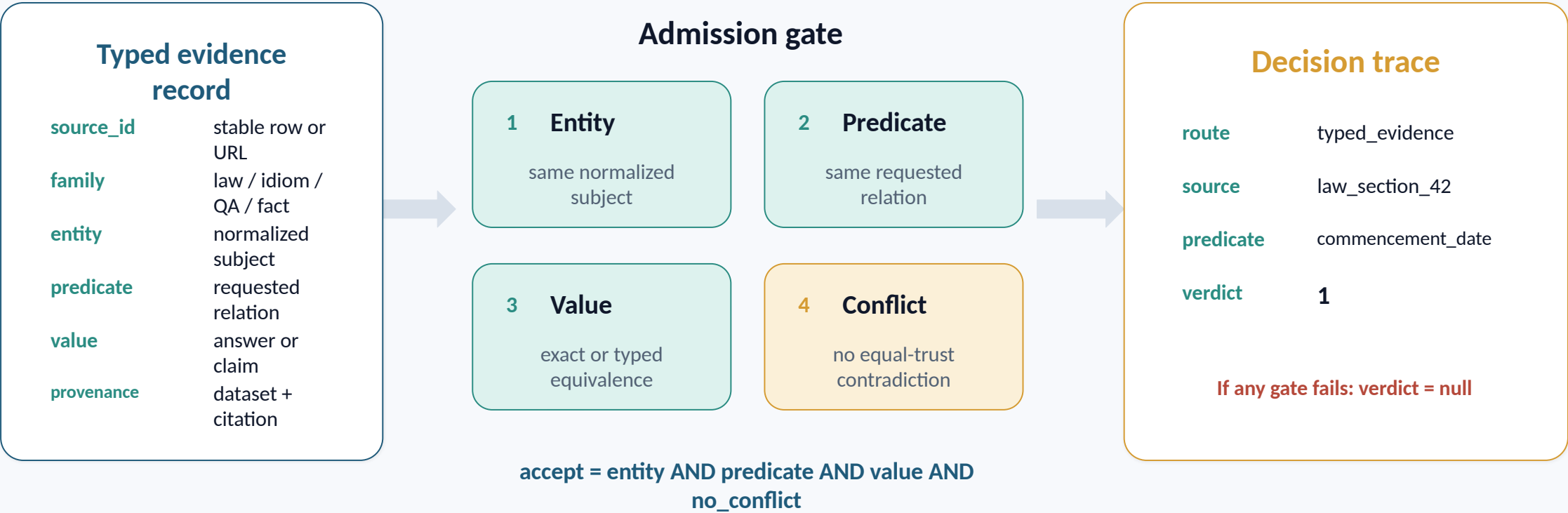
Fact check | exam grading | zero-bias check

Label 1 requires unanimous positive votes.

Qwen3.6-27B Q4_K_P | one-token grammar | fixed seeds and batches

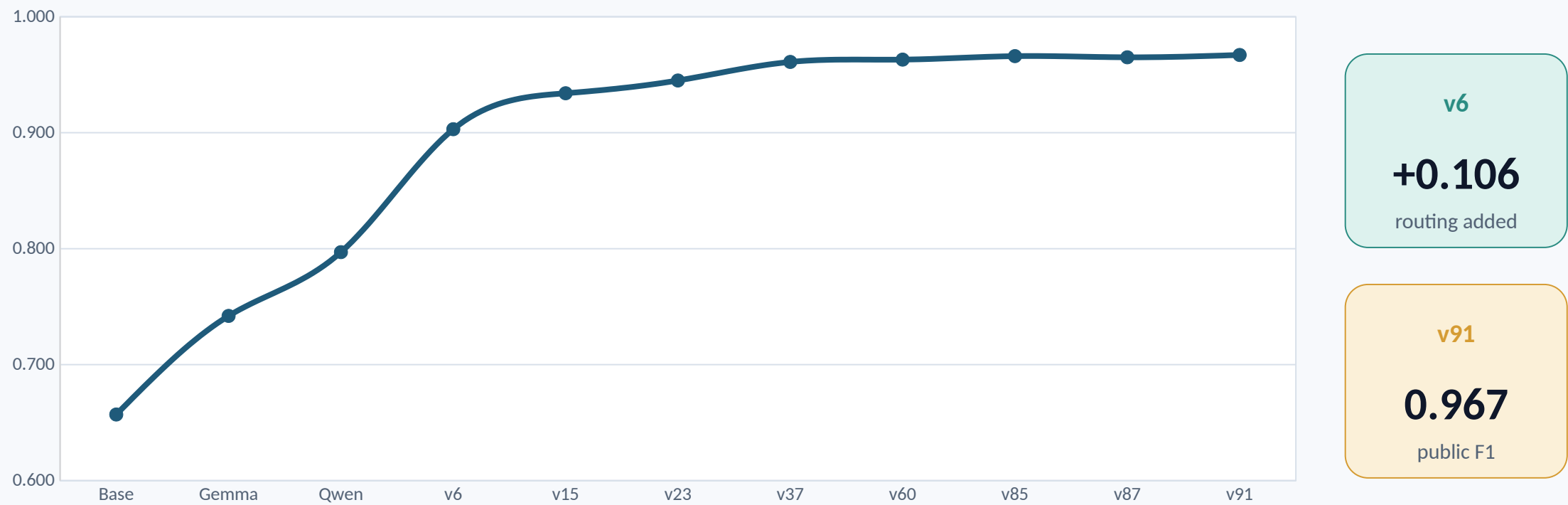
A source record was accepted only after four fields aligned

The trace preserved why a row was resolved, which source supported it, and which relation was tested.



Routing produced the largest score jump

The model helped. The architecture change helped more.



Several reasonable ideas made the score worse

The failed runs were useful because each exposed a different error source.

Experiment	Observed result	What it taught us
Prompt voting	0.851 local, 0.783 public	Overfit prompt behavior on 299 labels
TF-IDF verifier	0.826 out of fold	Missed no-context cultural facts
Raw QA retrieval	Relevant passage, wrong answer	Similarity confused neighboring relations
v87 fact edits	0.966 to 0.965 public	Plausible facts were not enough

After v87, no evidence record entered the router without an explicit source and relation.

[7] Rejected and scored experiment records in EXPERIMENT.log

The final result held across local, public, and private views

Each view answers a different question, so we report them separately.

0.9933

macro F1 on 299 released labels

1.0000

macro F1 on 225 deterministic controls

0.967

public macro F1, rank 5

0.982

private macro F1, rank 1

What we can claim

The final system generalized better than our public-only experiments suggested.

What we cannot claim

A private score does not prove every v91 rule will transfer to the broader Phase 2 sources.

Phase 2 completed in 3 h 52 min with 95.9% judged

Measured organizer-run inference on 5,000 rows using Kaggle Tesla T4 x2.

5,000

held-out rows

207

deterministic (4.14%)

4,793

Qwen judge (95.86%)

3:52:02

validated end to end

Measured milestones	Elapsed	Observation
Router complete	1:13.18	207 resolved; 4,793 abstained
Model ready	2:51.25	16.332 GiB Qwen across two T4 GPUs
Judge complete	3:52:01.91	13,747.50 s; 0.349 rows/s

42.97% of the 9-hour budget used

5:07:58 headroom

Phase 2 coverage fell from 93.72% to 4.14%

Exact Phase 1 lineage will cover less. The fallback must remain safe when coverage drops.

Web and news

Common Crawl, recent newspapers

Knowledge

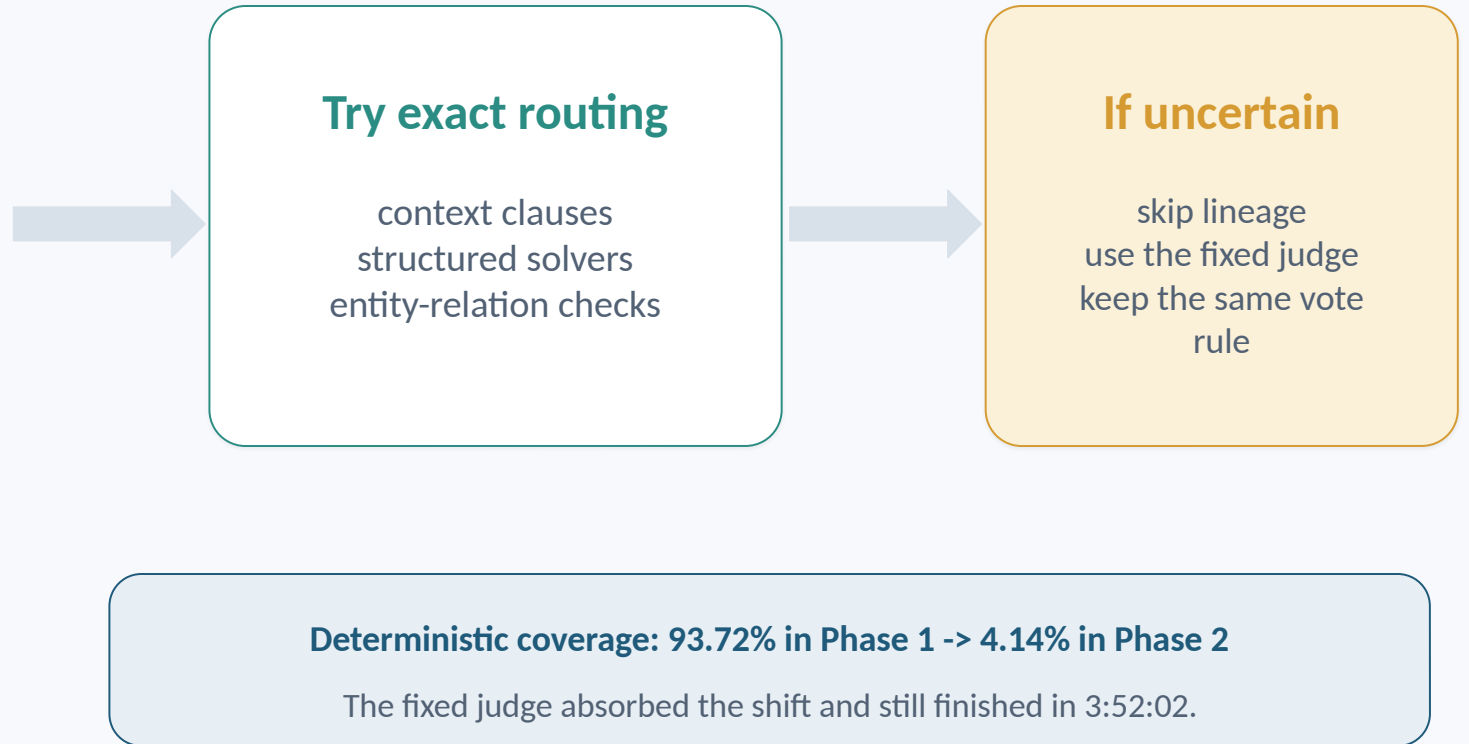
Wikipedia and Banglapedia

Civic and legal

government services and bdlaws

Education and culture

NCTB textbooks and literature



Be exact when possible. Abstain when necessary.

- 01 Task families need different evidence standards.

- 02 The requested relation matters more than semantic similarity.

- 03 **The fixed judge preserved compliance when source coverage collapsed.**

Questions

Private rank 1 | 0.982 macro F1 | 9-hour compliant